

Time-Eight Positive-Sojourn Classification for Arbitrary Two-State Unitary Coins

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Abstract

Fix the initial state $\phi_* = 2^{-1/2}(1, i)^T$ and Konno's interval convention for positive sojourn time, and condition an eight-step two-state quantum walk on return to the origin. We classify all coins $U \in U(2)$ for which the conditional law assigns probability $1/3$ to each of 2, 4, 6 and zero elsewhere. The complete raw solution set is

$$\left\{ D \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & z \\ 1 & -z \end{pmatrix} : D \text{ diagonal unitary, } |z| = 1 \right\}.$$

For every such coin the three nonzero unconditioned weights equal $1/128$. The return probability vanishes exactly for diagonal coins. Thus all solutions are complex Hadamard matrices in the entry-modulus sense. However, under the left-diagonal gauge and ordinary unitary basis changes that preserve the ordered chiralities and the fixed initial ray, the phase z is invariant; only $z = 1$ belongs to the restricted left-Hadamard orbit. Right column phases are not symmetries of the fixed experiment. If antiunitary basis changes are also admitted, the sole additional identification is $z \sim -\bar{z}$, so a continuum of classes remains.

1 Setting and result

Tamura and Yamagami proved that the time-eight uniformity condition is rigid inside the real rotation family and proposed its extension to general two-state unitary coins [1]. The point requiring care in that extension is that a familiar matrix equivalence need not preserve the fixed initial state, the ordered chirality projectors, or the positive half-line. We therefore classify the coins before quotienting and then test the relevant setup symmetries.

Our proof is an exact low-dimensional complex-matrix classification: a complete unitary normal form reduces the finite path sum to polynomial identities, after which every real branch and endpoint is resolved. As a methodological parallel only, Zeng, Liu, and Ratnavelu use exact rational certificates and equality classification in an unrelated low-dimensional Schatten-norm problem [2]. That preprint supplies no claim about quantum walks, unitary coins, sojourn statistics, or the rigidity problem considered here.

Write

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad P = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ c & d \end{pmatrix}.$$

The matrices P and Q represent a step to the left and right, respectively. For a word $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, +1\}^n$, let

$$x_0 = 0, \quad x_j = \sum_{r=1}^j \varepsilon_r, \quad M_{-1} = P, \quad M_{+1} = Q,$$

and set

$$A_U(\varepsilon) = M_{\varepsilon_n} \cdots M_{\varepsilon_1}.$$

The interval positive-sojourn count is

$$K(\varepsilon) = \sum_{j=0}^{n-1} \mathbf{1}\{x_j > 0 \text{ or } (x_j = 0 \text{ and } \varepsilon_{j+1} = +1)\}.$$

For a return word, define the coherent sum and its weight by

$$\Gamma_n(k; U) = \sum_{\substack{\varepsilon \in \{-1, +1\}^n \\ x_n = 0, K(\varepsilon) = k}} A_U(\varepsilon), \quad w_n(k; U) = \|\Gamma_n(k; U)\phi_*\|_2^2,$$

where $\phi_* = 2^{-1/2}(1, i)^T$. Put $R_n(U) = \sum_{k=0}^n w_n(k; U)$. The return-conditioned distribution $\mu_n(k; U) = w_n(k; U)/R_n(U)$ is defined only when $R_n(U) > 0$.

Let

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathcal{H}_{\text{left}} = \{DH : D \text{ diagonal unitary}\}.$$

Theorem 1 (Complete classification). *The condition*

$$R_8(U) > 0, \quad \mu_8(k; U) = \begin{cases} 1/3, & k \in \{2, 4, 6\}, \\ 0, & k \notin \{2, 4, 6\} \end{cases}$$

holds precisely for

$$U = DH_z, \quad H_z = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & z \\ 1 & -z \end{pmatrix}, \quad D \text{ diagonal unitary}, \quad |z| = 1. \quad (1)$$

For these coins,

$$w_8(2; U) = w_8(4; U) = w_8(6; U) = \frac{1}{128}, \quad R_8(U) = \frac{3}{128},$$

and every other weight is zero. On all of $U(2)$, $R_8(U) = 0$ if and only if U is diagonal.

Modulo left-diagonal multiplication and ordinary unitary internal-basis changes preserving the ordered chirality projectors and the ray of ϕ_ , the parameter $z \in \mathbb{T}$ is invariant. Hence $\mathcal{H}_{\text{left}}$ is exactly the slice $z = 1$, and coins with $z \neq 1$ form an exact counterfamily to rigidity with respect to that restricted orbit. If antiunitary internal-basis maps are admitted, the quotient acquires exactly the additional involution $z \mapsto -\bar{z}$.*

The first assertion says, equivalently, that the raw solutions are exactly the 2×2 unitary matrices whose four entries all have modulus $1/\sqrt{2}$. Thus “complex Hadamard” and “belonging to the restricted left-Hadamard orbit” must not be conflated.

2 Normal form and exact path sums

Lemma 2 (Complete normal form and left gauge). *Every $U \in U(2)$ has a representation*

$$U = DV(r, z), \quad V(r, z) = \begin{pmatrix} r & sz \\ s & -rz \end{pmatrix}, \quad r, s \geq 0, \quad r^2 + s^2 = 1, \quad |z| = 1, \quad (2)$$

where D is diagonal unitary. This includes $r = 0$ and $s = 0$. Moreover, every time-eight return weight is unchanged by the factor D .

Proof. Independent row phases make the first column of U equal to $(r, s)^T$ with $r, s \geq 0$. Its unit orthogonal complement is $z(s, -r)^T$ for some $|z| = 1$, which proves (2) without dividing by r or s . If $D = \text{diag}(\lambda, \nu)$, then the step matrices of DU are λP and νQ . Every eight-step return word contains four left and four right steps, so its amplitude is multiplied by the same phase $\lambda^4 \nu^4$. The coherent sums acquire that common phase and their weights are unchanged. $\square$

It remains to calculate for $V(r, z)$. Put

$$x = r^2, \quad y = s^2 = 1 - x, \quad u = rs, \quad h = \text{Im } z$$

and introduce the real polynomials

$$\begin{aligned} f &= y^3 - 6xy^2 + 6x^2y - x^3, \\ A &= -y^2 + 3xy - x^2, & B &= y^2 - 5xy + 3x^2, \\ C &= -3y^2 + 5xy - x^2, & M &= -2y^2 + 4xy - x^2, \\ N &= -y^2 + 4xy - 2x^2. \end{aligned}$$

Lemma 3 (The five coherent sums). *The $70 = \binom{8}{4}$ return words split into fourteen words in each of $k = 0, 2, 4, 6, 8$, with no odd sojourn class, and their coherent sums are*

$$\Gamma_8(0) = f \begin{pmatrix} 0 & 0 \\ rsz^3 & s^2z^4 \end{pmatrix}, \quad (3)$$

$$\Gamma_8(2) = \begin{pmatrix} u^2Az^4 & uyAz^5 \\ uyBz^3 & u^2Cz^4 \end{pmatrix}, \quad (4)$$

$$\Gamma_8(4) = \begin{pmatrix} u^2Mz^4 & uyNz^5 \\ -uyNz^3 & u^2Mz^4 \end{pmatrix}, \quad (5)$$

$$\Gamma_8(6) = \begin{pmatrix} u^2Cz^4 & -uyBz^5 \\ -uyAz^3 & u^2Az^4 \end{pmatrix}, \quad (6)$$

$$\Gamma_8(8) = f \begin{pmatrix} s^2z^4 & -rsz^5 \\ 0 & 0 \end{pmatrix}. \quad (7)$$

Proof. This is a finite exact enumeration. For completeness, it can be reproduced without symbolic factorization as follows. Start with the identity matrix at time 0, position 0, and sojourn count 0. At each state (t, j, k) , left-multiply its accumulated matrix by P for the transition $j \mapsto j - 1$, and by Q for $j \mapsto j + 1$; increment k exactly when $j > 0$, or when $j = 0$ and the chosen transition is to the right. Collect equal triples after each step and retain position 0 at time 8. Substitution of

$$P = \begin{pmatrix} r & sz \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ s & -rz \end{pmatrix}$$

and collection of monomials gives (3)–(7). This recurrence performs only additions and multiplications in $\mathbb{Z}[r, s, z]$; an accompanying standard-library script carries out the 70-word expansion exactly. $\square$

For real X, Y and $|z| = 1$,

$$|X + izY|^2 = X^2 + Y^2 - 2XY \text{Im } z. \quad (8)$$

Define

$$W = \frac{1}{2} \left(u^4 A^2 + u^2 y^2 A^2 + u^2 y^2 B^2 + u^4 C^2 \right).$$

Applying (8) to the five matrices yields

$$w_8(0) = \frac{y f^2}{2} (1 - 2uh), \quad w_8(8) = \frac{y f^2}{2} (1 + 2uh), \quad (9)$$

$$w_8(2) = W - u^3 y h (A^2 + BC), \quad w_8(6) = W + u^3 y h (A^2 + BC), \quad (10)$$

$$w_8(4) = u^4 M^2 + u^2 y^2 N^2. \quad (11)$$

The decisive polynomial identities are

$$f = (1 - 2x)(1 - 7x + 7x^2), \quad (12)$$

$$A^2 + BC = 2(2x - 1)f, \quad (13)$$

$$W - w_8(4) = x^2 y^3 (2x - 1)^2. \quad (14)$$

They follow by substituting $y = 1 - x$ and collecting powers of x ; in particular, they are identities in $\mathbb{Z}[x]$, not numerical fits.

3 Solving all branches

Proof of Theorem 1: raw classification. By Lemma 2, it suffices to use the chart (2). The two forbidden endpoint weights obey

$$w_8(0) + w_8(8) = y f^2. \quad (15)$$

If both vanish, then either $y = 0$ or $f = 0$. On the branch $y = 0$ we have $s = 0$, and every matrix in (3)–(7) is zero. Hence $R_8 = 0$, so the conditional law is undefined and this branch cannot be a solution.

Suppose now that $y > 0$ and $f = 0$. Since $f(0) = 1$, we also have $x > 0$. Equation (13) removes the z -dependent terms in (10), while (14) gives

$$w_8(2) = w_8(6) = w_8(4) + x^2 y^3 (2x - 1)^2. \quad (16)$$

Equality of the three target weights therefore forces $x = 1/2$; no division by an endpoint parameter has been used. Conversely, exact substitution at $x = y = 1/2$ gives

$$w_8(2) = w_8(4) = w_8(6) = \frac{1}{128}$$

for every $|z| = 1$, while (9) and the absence of odd classes give zero elsewhere. Thus $R_8 = 3/128$, and Lemma 2 gives exactly (1). The two other roots $(7 \pm \sqrt{21})/14$ of $1 - 7x + 7x^2$ fail equality because the last term of (16) is strictly positive.

It remains to identify the zero-return locus. We have already seen that $y = 0$ makes every return amplitude vanish. If $y > 0$ and $f \neq 0$, then (15) is positive. If $y > 0$ and $f = 0$, then $x > 0$ and $u > 0$. Were $w_8(4) = 0$, equation (11) would force $M = N = 0$. But

$$M - N = x^2 - y^2 = 2x - 1,$$

whereas at $x = 1/2$ one has $M = N = 1/4$, a contradiction. Hence $R_8 > 0$ there. Therefore $R_8 = 0$ exactly when $s = 0$, which in the normal form is exactly the locus of diagonal unitary coins. $\square$

4 Equivalences of the fixed experiment

The left-diagonal action in Lemma 2 is a genuine pathwise symmetry. On the balanced locus, the remaining phase is

$$z = \frac{b}{a},$$

so left row phases do not change it and $\mathcal{H}_{\text{left}}$ is precisely $z = 1$.

Consider next an ordinary unitary change of internal basis that is required to preserve the two ordered chirality projectors. It must commute with both $\text{diag}(1, 0)$ and $\text{diag}(0, 1)$, hence is diagonal. If it also preserves the ray of ϕ_* , its two diagonal phases agree. It is therefore scalar, and conjugation by it acts trivially on the coin. No further unitary-basis identification of z is available.

Right diagonal multiplication is not a symmetry of this fixed setup. For an exact witness, take $x = 1/4$ in (2). The coins with $z = 1$ and $z = i$ differ by right multiplication by $\text{diag}(1, i)$, but

$$w_8(0; z = 1) = \frac{75}{8192}, \quad w_8(0; z = i) = \frac{75}{8192} \left(1 - \frac{\sqrt{3}}{2}\right).$$

Thus the fact that right phases happen not to alter the target law on the balanced solution locus cannot promote them to symmetries of the experiment.

For completeness, suppose antiunitary Wigner changes of internal basis are allowed. An antiunitary preserving the ordered chirality projectors is a diagonal unitary followed by complex conjugation K . Imposing preservation of the ray of ϕ_* leaves, up to an overall phase,

$$T = JK, \quad J = \text{diag}(1, -1).$$

It fixes ϕ_* and induces

$$U \mapsto J\bar{U}J, \quad \Gamma_n(k; U) \mapsto J\overline{\Gamma_n(k; U)}J.$$

Since $J\phi_* = \overline{\phi_*}$, every weight is preserved. Returning the transformed balanced coin to the left-gauge normal form sends $z \mapsto -\bar{z}$. This proves that the optional quotient is $\mathbb{T}/(z \sim -\bar{z})$, not a point. For example, the class through $z = 1$ also contains $z = -1$, while $z = i$ remains outside it. This completes the proof of Theorem 1.

Computational and AI assistance disclosure

The finite path enumeration and the polynomial identities were checked using a standard-library Python script with exact integer and rational arithmetic; floating-point sampling is not used in the proof. A separate verification implementation also reconstructed the path sums and identities during mathematical review. OpenAI Codex assisted with exact enumeration, proof organization, adversarial branch checking, and manuscript preparation. The author is responsible for the mathematical claims and for compliance with any venue-specific authorship and disclosure rules.

References

- [1] Shunya Tamura and Tomoki Yamagami. Hadamard rigidity of positive sojourn time distributions for rotation coins, 2026. arXiv:2609.05033v1 [quant-ph], submitted 4 September 2026; DOI: 10.48550/arXiv.2609.05033.
- [2] Zijian Zeng, Houde Liu, and Kurunathan Ratnavelu. A counterexample to the Tang–Zhang Schatten norm conjecture and sharp positive results. *Preprints*, 2026. Version 1, posted 17 August 2026; DOI: 10.20944/preprints202608.1067.v1; not peer reviewed.